Supplementary Material: Auditing Selection Purity and Transfer Utility under Joint Model and Data Heterogeneity

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Scope

This supplement retains local-adaptation and natural weak-label evidence that bounds, but does not add server endpoints to, the main study. We preserve the stated development-only, post-observation, and reused-test evidence labels; relocation outside the main-paper page limit does not upgrade them.

Closest-Work Boundary

Table 1 states the nearest-method boundary without claiming a new optimizer or native-setting dominance.

Complete Legacy Server Endpoints

Table 2 preserves the task-specific discovery endpoints that contextualize the main audit.

Full Server Configuration

All server audits use Qwen2.5-0.5B, SmolLM2-360M, and TinyLlama-1.1B clients with a Qwen2.5-1.5B server. LoRA targets query/value projections with rank 4, scale 8, and zero dropout. Each run uses two rounds, 40 client and 40 server steps per round, batch size 8, client learning rate 10^{-3}, server learning rate 10^{-4}, anchor weight 1, and distillation weight 0.25. Clients transmit the uniform mean of canonical verbalizer probabilities; no model parameter is aggregated across architectures.

The four arms are: A, public-anchor-only server training; B, uniform transfer from clean clients; C, uniform transfer from the joint severe mixture; and D, the complete severe routed intervention. AG News uses 256 anchors, 800 development rows, and the 7,600-row official test. TREC uses 72 anchors, 72 development questions, three 120-question client pools, and the 500-question official test. IMDb uses 128 anchors, 1,000 development reviews, three 256-example client pools, and all 25,000 official test reviews. Each severe IMDb client contains 103 clean, 102 conflict, and 51 off-task examples.

The IMDb secondary addendum fixes $K = 128$, 64 rows per claimed label, the severe source mixture, optimizer steps, and all server settings. Random-K samples only from target-eligible rows; routed-K uses the frozen routing order; maximum-purity-K contains 103 clean and 25 conflict rows. The last arm is an audit-only attainable upper control, not a clean-only method.

Post-hoc IMDb Selector-Component Diagnostic

After the IMDb primary and matched-K results were observed, we ran a post-hoc reused-test diagnostic that replaces the full routed-K score with one component at a time while preserving $K = 128$, 64 examples per claimed label, the severe mixture, models, anchors, optimizer settings, and server training budget. The new 30 records cover ten seeds and three arms: gradient-only, loss-only, and margin-only. The existing verified routed-K matched-size records serve as the full-router reference.

Mean accuracies are 91.46% for full routed-K, 91.30% for gradient-only, 91.50% for loss-only, and 91.36% for margin-only. The three post-hoc paired contrasts use one Holm family: full-minus-gradient is $+0.16$ pp (95% CI $[+0.00, +0.32]$, Holm $p = .135$); full-minus-loss is -0.04 pp ($[-0.17, +0.09]$, Holm $p = .505$); and full-minus-margin is $+0.10$ pp ($[-0.00, +0.21]$, Holm $p = .135$). Audited clean fractions are 71.0%, 59.4%, 67.3%, and 71.6%, respectively. Thus the diagnostic does not isolate a statistically detectable single-component replacement gap after multiplicity correction, and it does not change the primary IMDb gate or the matched-K secondary family.

Compatible Co-Teaching Control

We compare naive LoRA to two-peer Co-Teaching on frozen AG News development mixtures. Co-Teaching receives the true total contamination fraction as its forget rate, and peer probabilities are ensembled at evaluation. This is an oracle-rate compatible-model control, not a heterogeneous server baseline.

No contrast survives Holm correction over the six contaminated comparisons (all adjusted $p = 1.0$), and seed-wise effects change sign in five settings. The jointly contaminated TinyLlama setting also lies outside Co-Teaching's original finite-label-noise scope. Concurrently scheduled wall-time ratios of 2.84–$9.32\times$ are environment-specific diagnostics.

Natural Weak Labels

Hausa and Yoruba topic-classification corpora provide naturally occurring weak labels paired with released clean annotations (Hedderich et al. 2020). The weak/clean disagreement rates are 50.37% for 2,045 Hausa training headlines and

Work	Main focus	Boundary relative to this audit
Selective-FD (Shao, Wu, and Zhang 2024)	Filters misleading client-side distillation signals	Does not run four contaminated-client arms alongside matched-K and maximum-purity controls.
FedID (Ma et al. 2023)	Interactive distillation for pretrained language models in federated settings	Studies heterogeneous federated distillation without auditing clean/conflict/off-task purity against server utility.
FedHDS (Qin et al. 2025)	Representative data selection for federated LLM instruction tuning	Addresses data-efficient local tuning, whereas our selection is an audit intervention on transferred canonical probabilities.
FedFD (Li et al. 2025)	Alignment for unstable heterogeneous logit ensembles	Establishes that heterogeneous logit ensembles can fail; our contribution is evidence-labelled failure analysis under internal contamination.

Table 1: Closest-work boundary. The paper claims a narrow audit contribution, not a new optimizer or dominance over these methods in their native settings.

Task	Method	Accuracy	Macro-F1	Score ECE
AG News	Anchor-only	84.34 ± 0.62	84.28	0.054
AG News	Routed	83.78 ± 0.54	83.67	0.016
TREC	Anchor-only	80.72 ± 1.41	77.65	0.068
TREC	Routed	81.12 ± 0.67	78.38	0.031
TREC post	Anchor-only	81.76 ± 2.01	78.39	0.060
TREC post	Clean uniform	81.56 ± 1.55	79.01	0.035
TREC post	Severe uniform	83.14 ± 0.77	80.80	0.099
TREC post	Severe routed	81.72 ± 1.38	79.43	0.039

Table 2: Verified AG News and TREC server endpoints retained from the discovery stages. Accuracy is mean $\pm$ SD; macro-F1 and score ECE are seed means. TREC post rows reuse the already accessed test with ten new seeds.

Client	Conflict 40%	Joint 40%/20%
Qwen	59.12 / 60.92 (+1.79)	53.71 / 54.50 (+0.79)
SmolLM2	65.83 / 68.46 (+2.63)	48.00 / 47.37 (-0.63)
TinyLlama	59.25 / 68.54 (+9.29)	54.92 / 37.25 (-17.67)

Table 3: Three-seed AG News development accuracy: naive / Co-Teaching (percentage-point change).

33.28% for 1,340 Yoruba headlines. The corpus-level rate is supplied as Co-Teaching's oracle forget rate; per-example clean labels never enter model input, gradients, or selection scores. Only the clean development splits are evaluated.

Across three seeds, Hausa naive/Co-Teaching accuracies are 44.94/33.33 for Qwen, 45.29/28.16 for SmolLM2, and 47.82/28.74 for TinyLlama, giving changes of -11.61, -17.13, and -19.08 pp. The frozen gate fails and the clean test split remains unused. Yoruba changes are $+0.53$, $+2.65$, and -4.23 pp; all intervals cross zero, so its gate also fails and its test split is not evaluated. Co-Teaching raises Yoruba selected clean-label agreement from 66.72% to 78.2%–80.6% without a reliable accuracy gain. This is a local counterexample to treating selection purity as sufficient for utility, not a server-transfer result. In Hausa, weak labels match the released clean labels on 49.63% of training examples, while selected agreement ranges from 45.3%–50.9%. Its concurrently scheduled Co-Teaching runs cost 1.69–1.80$\times$ the naive wall time; these are environment-specific diagnostics.

Open-Answer Teacher Ranking

GSM8K supplies target questions and SQuAD supplies off-task QA (Cobbe et al. 2021; Rajpurkar et al. 2016). A teacher is trained only on the public anchor and ranks supplied numerical answers by token-normalized answer loss. For a supplied pair $z = (x, \widetilde{y})$, its score is

$$s_T(z) = -|\widetilde{y}|^{-1} \log p_{f_T}(\widetilde{y} \mid x).$$

After a numerical-interface check, the client retains the K_{local} highest-score examples and performs the same optimizer-step budget as the unfiltered arm. No teacher parameter, hidden state, validation label, or test example enters client training. On the official GSM8K test, Qwen severe-mixture exact match rises from 3.49% EM to 6.82% ($+3.34$ pp); clean exact match rises from 6.62% to 7.91%. This supports a narrow local Qwen endpoint, not beneficial federated transfer. The client families are documented in their model reports (Qwen Team 2025; Ben Allal et al. 2025; Zhang et al. 2024).

Client	Condition	Naive	Teacher	Gain
SmolLM2	clean	4.55	4.90	+0.35
SmolLM2	severe	2.58	4.17	+1.59
TinyLlama	clean	3.66	4.25	+0.58
TinyLlama	severe	1.42	3.44	+2.02

Table 4: Post-observation non-Qwen GSM8K extension. The two severe hypotheses form a Holm family.

All six seed-level severe differences are positive, and both clean safety gates hold. Nevertheless, both severe confidence intervals cross zero and both Holm-adjusted tests have $p = 0.0714$; the declared cross-family claim fails. On development data, teacher ranking exceeds matched numeric-random and client small-loss selectors by $+0.78/+0.98$ pp for SmolLM2 and $+1.56/+1.50$ pp for TinyLlama. These post-hoc controls cannot repair the failed official cross-family gate.

The descriptive curve gate passes, but none of nine family-by-severity tests survives joint Holm correction. Matched development selectors support a teacher-signal interpretation without repairing the failed official cross-family gate. A matched 64-question diagnostic gives $1.52\times$ and $1.58\times$ wall-time ratios for SmolLM2 and TinyLlama; serial scoring of 256 candidates takes approximately 8.1 seconds and peaks at 5.57 GiB allocated GPU memory on the recorded host.

Artifact Reconstruction and Failure Policy

The reconstruction path is content addressed rather than inferred from a paper table. Each server study has a frozen protocol and manifest recording the task specifications, source-file hashes, model-weight hashes, data partitions, seed set, method matrix, and analysis rule. Protocol histories also retain pre-evaluation instrumentation repairs instead of silently rewriting their chronology. Development and official evaluation use separate phase directories, seed ranges, and output names. Where a development phase must not access a test split, a nonexistent sentinel is part of the verifier's contract. Thus a development artifact cannot be relabelled as an official record merely by moving or renaming it.

Reconstruction begins from the per-run JSON records, not from rounded values in the manuscript. A phase verifier checks the expected seed–method Cartesian product, record identities, frozen protocol and configuration fields, evaluation counts, finite metrics, and method-specific invariants. The IMDb matched-size analysis additionally verifies its fixed cardinality, claimed-label quotas, mixture identity, and a separate conditional implementation relationship receipt for the routed original and routed-K arms. The paper-asset generator first requires the exact complete matrices and positive verifier certificates. Only then does it load the raw JSON, recompute arm means and sample standard deviations, paired differences and Student-t intervals, the frozen paired tests and Holm adjustment, exact sign-flip sensitivity where declared, and audited selection composition. A single machine-readable statistics file is then used to emit the TeX macros, tables, English results block, and vector figure. These files are built in a temporary staging directory and replace the published asset set only after figure compilation succeeds.

The routed relationship verifier was corrected after all 70 IMDb official records already existed. The verifier frozen at official start (`2c8678f7...`) incorrectly required unconditional whole-record identity, even when the original conditional gate retained all 256 examples and the matched protocol deliberately forced $K = 128$. The repaired verifier (`5518b177...`) checks exact identity only before a local gate fork, records the original gate decision as the only authorized first divergence, and then checks each arm's frozen branch invariants. It partitions the 60 client-rounds into 55 exact pre-fork triggered rounds, four first authorized gate divergences, and one post-local-fork round that is not relationship-eligible; six of ten seed trajectories are fully identical. No raw JSON, seed, arm, metric, statistic, protocol, or runner changed, and this receipt is separate from the three scientific IMDb gate clauses.

Failure is therefore explicit and fail-closed. A missing or extra record, an identity mismatch, a non-finite metric, a verifier rejection, a failed conditional relationship receipt, or failed figure compilation prevents publication of a new asset set unless a verifier defect is disclosed, patched, and regression-tested without changing records or reported statistics. Aborted and invalidated attempts may remain as timestamped failure logs for audit, but they are not members of the verified matrix and cannot enter a summary. Partial official results may be observed solely to diagnose process health; they do not authorize seed changes, stopping, hyperparameter updates, new contrasts, or manuscript numbers. Official claims are generated only after the full frozen matrix passes its verifier. These checks establish the reported artifact path; they are not a claim that every legacy or optional test in the broader repository succeeds in the present environment.

Evidence Boundary

The GSM8K Qwen result is the only positive official local endpoint in this supplement. The non-Qwen extension follows observation of the Qwen phase on the same official test, the severity curve is post-primary, and both natural weak-label studies are development-only. None of these studies establishes a general federated algorithm, privacy property, or task-population effect.

Client	Clean		Medium (20/10)		Severe (40/20)	
	Naive	Teacher	Naive	Teacher	Naive	Teacher
Qwen	6.38	6.84 (+0.46)	5.24	7.85 (+2.60)	1.69	5.76 (+4.07)
SmolLM2	3.68	2.80 (-0.88)	2.15	3.39 (+1.24)	1.46	2.47 (+1.01)
TinyLlama	2.99	2.93 (-0.07)	2.02	2.44 (+0.42)	1.07	1.73 (+0.65)

Table 5: Disjoint 1,024-example post-primary severity curve. Values are mean exact match; parentheses show teacher-minus-naive percentage-point changes.

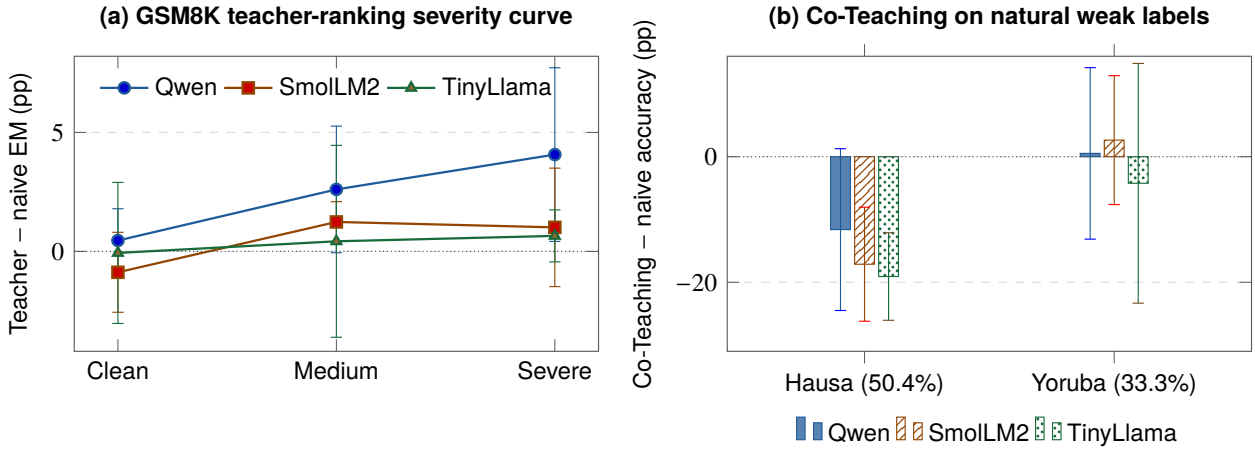


Figure 1: Paired mean changes with descriptive 95% Student-t intervals over three seeds. Left: GSM8K teacher-ranking severity curve. Right: Hausa and Yoruba oracle-rate Co-Teaching effects. Intervals describe seed variation; the pre-specified gates described above remain failed.

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